Online Examination Report

Date:	22.08.2014
Attempts:	1
Examinee:	Echel, Isaac
Date of birth:	
Subject:	080- Principles of flight
Result:	87,5 %
Time used:	00:25:55 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	209	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
2	218	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
3	202	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
4	180	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
5	187	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
6	233	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
7	214	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
8	230	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
9	195	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
10	204	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
11	149	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
12	208	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
13	207	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
14	220	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
15	226	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
16	20	81.1.1.1	Laws and definitions	1,00	1,00
17	48	81.1.1.1	Laws and definitions	1,00	1,00
18	60	81.1.1.3	Aerodynamic forces and moments on aerofoils	1,00	1,00
19	61	81.1.1.4	Shape of an aerofoil section	1,00	1,00
20	23	81.1.2.4	Centre of pressure and aerodynamic centre	1,00	1,00
21	4	81.1.4.2	Induced drag	1,00	1,00
22	78	81.1.12.1	Ice and other contaminants	1,00	1,00
23	36	81.2.2	Shock waves	1,00	1,00
24	59	81.2.2.2	Oblique shock waves	1,00	1,00
25	18	81.2.3.2	Effect on lift	1,00	1,00
26	42	81.2.3.4	Effect on pitching moment	0,00	1,00
27	45	81.2.4	Buffet onset	1,00	1,00
28	87	81.4.3.3	Neutral point	1,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	15	81.4.4.4	Factors affecting static directional stability	1,00	1,00
30	5	81.4.5.5	Factors affecting static lateral stability	0,00	1,00
31	21	81.5.2.4	Location of centre of gravity	1,00	1,00
32	77	81.5.8.3	Stabiliser trim	1,00	1,00
33	6	81.6.1.1	Flutter	1,00	1,00
34	50	81.7.1.3	Fixed and variable pitch/constant speed	1,00	1,00
35	73	81.7.1.4	Propeller efficiency versus speed	1,00	1,00
36	8	81.7.4.2	Gyroscopic precession	0,00	1,00
37	24	81.8.1.2	Straight steady climb	1,00	1,00
38	31	81.8.1.4	Straight steady glide	1,00	1,00
39	68	81.8.1.5	Steady coordinated turn	1,00	1,00
40	92	81.8.3	Particular points on the polar curve	1,00	1,00
			Total:88%	**35,00**	**40,00**

Attachment 2: List of result by topic

Licence
Date: 22.08.2014
Examinee: Echel, Isaac
Location: Uganda Civil Aviation Authority

Your Result: 35,00 from 40,00 Points (87,50 %)

Topics	Ammount of Questions	Points achieved	Maximum Points	Percent
81. PRINCIPLES OF FLIGHT AEROPLANE	40	35,00	40,00	87,50
81.1. SUBSONIC AERODYNAMICS	7	7,00	7,00	100,00
81.2. HIGH SPEED AERODYNAMICS	5	4,00	5,00	80,00
81.4. STABILITY	3	2,00	3,00	66,67
81.5. CONTROL	2	2,00	2,00	100,00
81.6. LIMITATIONS	1	1,00	1,00	100,00
81.7. PROPELLERS	3	2,00	3,00	66,67
81.8. FLIGHT MECHANICS	4	4,00	4,00	100,00
Total	**40**	**35,00**	**40,00**	**87,50**

1

Unless adequate speed control is maintained during the turn to base and the final approach for a landing into the wind, which would most likely occur if a steep wind gradient existed?

- [x] **The desired landing spot would be undershot or the glider would stall.**
- [ ] **The wingtip on the outside of the turn would stall before the wingtip on the inside of the turn.**
- [ ] **The airspeed on final approach would increase, causing the glider to overshoot the desired landing spot.**

ID:	209(LTS:226390, LTM:209)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	1,00

2

The practice of allowing the ground crew to lift a balloon into the air is

- [x] **unsafe because it can lead to a sudden landing at an inopportune site just after lift-off.**
- [ ] **considered to be good practice, particularly when obstacles must be cleared shortly after lift-off.**
- [ ] **a safe way to reduce stress on the envelope.**

ID: 218(LTS:226399, LTM:218)
Syllabus reference: 81. PRINCIPLES OF FLIGHT AEROPLANE
Max. Points 1,00
Points 1,00

3

What consideration should be given in the choice of a towplane for use in aerotows?

- ☐ Gross weight of the glider to be towed.
- ☒ Stall speed of the towplane.
- ☐ Towplane's low-wing loading and low-power loading.

ID: 202(LTS:226383, LTM:202)
Syllabus reference: 81. PRINCIPLES OF FLIGHT AEROPLANE

Max. Points 1,00
Points 1,00

4

Which type of flap is characterized by large increases in lift coefficient with minimum changes in drag?

- [x] **Fowler.**
- [ ] **Split.**
- [ ] **Slotted.**

ID:	180(LTS:226363, LTM:180)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	1,00

5

Which characteristic of a spin is not a characteristic of a steep spiral?

- ☐ High rate of rotation.
- ☐ Rapid loss of altitude.
- ☒ Stalled wing.

ID:	187(LTS:226369, LTM:187)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	1,00

6

After being handed off to the final approach controller during a 'no-gyro' surveillance or precision approach, the pilot should make all turns

- ☐ **based upon the groundspeed of the aircraft.**
- ☒ **one-half standard rate.**
- ☐ **standard rate.**

ID:	233(LTS:226413, LTM:233)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	1,00

7

What is the suggested speed to fly when passing through lift with no intention to work the lift?

- [ ] **Best glide speed.**
- [ ] **Best lift/drag speed.**
- [x] **Minimum sink speed.**

ID: 214(LTS:226395, LTM:214)
Syllabus reference: 81. PRINCIPLES OF FLIGHT AEROPLANE

Max. Points 1,00
Points 1,00

8

The wing of a weight shift aircraft twists so that the angle of attack

- ☐ from the center of the wing to the wing tip is variable and can be adjusted by the pilot in flight to optimize performance.
- ☐ changes from a high angle of attack at the center of the wing, to a low angle of attack at the tips.
- ☒ changes from a low angle of attack at the center of the wing, to a high angle of attack at the tips.

ID:	230(LTS:226410, LTM:230)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	0,00

9

When an aircraft's forward CG limit is exceeded, it will affect the flight characteristics of the aircraft by producing

- [ ] very light elevator control forces which make it easy to inadvertently overstress the aircraft.
- [x] higher stalling speeds and more longitudinal stability.
- [ ] improved performance since it reduces the induced drag.

ID: 195(LTS:226376, LTM:195)
Syllabus reference: 81. PRINCIPLES OF FLIGHT AEROPLANE

Max. Points 1,00
Points 1,00

10

How can excessive towline slack that is allowed to develop during a glider tow be eliminated?

- ☐ Increase pitch attitude until towline becomes taut.
- ☒ Yaw the nose to one side with rudder while keeping the wings level with ailerons.
- ☐ Execute a shallow banked coordinated turn to either side.

ID: 204(LTS:226385, LTM:204)
Syllabus reference: 81. PRINCIPLES OF FLIGHT AEROPLANE

Max. Points 1,00
Points 1,00

11

At what bank angle will the resultant of gravity and centrifugal force equal twice a glider's weight?

- [x] **60 degree.**
- [ ] **45 degrees.**
- [ ] **30 degrees.**

ID: 149(LTS:226361, LTM:149)
Syllabus reference: 81. PRINCIPLES OF FLIGHT AEROPLANE
Max. Points 1,00
Points 1,00

12

During a winch launch, which factor would most likely result in pitch oscillations?

- ☐ **Winching speed too slow.**
- ☒ **Insufficient up-elevator control.**
- ☐ **Winching speed too fast.**

ID:	208(LTS:226389, LTM:208)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	1,00

13

At what point during an autotow should the glider pilot establish the maximum pitch attitude for the climb?

- ☐ **100 feet above the ground.**
- ☒ **200 feet above the ground.**
- ☐ **Between 300 and 400 feet above the ground.**

14

What is one procedure for relighting the burner while in flight?

- ☐ **Close the tank valves, vent the fuel lines, reopen the tank valves, and light the pilot light.**
- ☒ **Open another tank valve, open the blast valve, and light the main jet using reduced flow.**
- ☐ **Open the blast valve full open and light the pilot light.**

ID:	220(LTS:226401, LTM:220)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	1,00

15

During a wing stall, the wing tips of a weight shift aircraft are

☒ ineffective for stall recovery.

☐ effective only when combined with maximum engine output.

☐ effective for stall recovery.

ID: 226(LTS:226407, LTM:226)
Syllabus reference: 81. PRINCIPLES OF FLIGHT AEROPLANE

Max. Points 1,00
Points 0,00

16

The SI unit of measurement for density is:

- ☐ **psi.**
- ☐ **bar.**
- ☒ **kg/m^3.**
- ☐ **kg/cm^2.**

ID:	20(LTS:226232, LTM:20)	Max. Points	1,00
Syllabus reference:	81.1.1.1 Laws and definitions	Points	1,00

17

Considering subsonic incompressible airflow through a Venturi, which statement is correct?

I. The static pressure in the undisturbed airflow is lower than in the throat.
II. The speed in the undisturbed airflow is lower than in the throat.

- ☐ I is correct, II is incorrect.
- ☐ I is incorrect, II is incorrect.
- ☐ I is correct, II is correct.
- ☒ I is incorrect, II is correct.

ID:	48(LTS:226260, LTM:48)	Max. Points	1,00
Syllabus reference:	81.1.1.1 Laws and definitions	Points	1,00

18

The angle of attack of an aerofoil section is defined as the angle between the:

- ☐ local airflow and the mean camber line.
- ☐ undisturbed airflow and the mean camber line.
- ☒ undisturbed airflow and the chord line.
- ☐ local airflow and the chord line.

ID: 60(LTS:226272, LTM:60)
Syllabus reference: 81.1.1.3 Aerodynamic forces and moments on aerofoils

Max. Points 1,00
Points 1,00

19

An aerofoil is cambered when:

- ☐ the upper surface of the aerofoil is curved.
- ☐ the chord line is curved.
- ☒ the line, which connects the centres of all inscribed circles, is curved.
- ☐ the maximum thickness is large compared with the length of the chord.

ID: 61(LTS:226273, LTM:61)
Syllabus reference: 81.1.1.4 Shape of an aerofoil section

Max. Points 1,00
Points 1,00

20

Assuming no flow separation and no compressibility effects the location of the centre of pressure of a symmetrical aerofoil section:

- [x] is at approximately 25% chord irrespective of angle of attack.
- [ ] is at approximately 50% chord irrespective of angle of attack.
- [ ] moves backward when the angle of attack decreases.
- [ ] moves forward when the angle of attack decreases.

ID:	23(LTS:226234, LTM:23)	Max. Points	1,00
Syllabus reference:	81.1.2.4 Centre of pressure and aerodynamic centre	Points	1,00

21

Which statement concerning the local flow pattern around a wing is correct?

- ☐ Vortex generators on the wing partially block the spanwise flow over the wing
- ☒ By fitting winglets to the wing tip,reduces induced drag.
- ☐ Sweepback reduces drag since, because the span increases.
- ☐ Slat extension, at a constant angle of attack and normal extension speeds, will increase the lift coefficient,

ID:	4(LTS:226216, LTM:4)	Max. Points	1,00
Syllabus reference:	81.1.4.2 Induced drag	Points	1,00

22

The effects of very heavy rain (tropical rain) on the aerodynamic characteristics of an aeroplane are:

☐ decrease of CLMAX and decrease of drag.

☐ increase of CLMAX and decrease of drag.

☐ increase of CLMAX and increase of drag.

☒ decrease of CLMAX and increase of drag.

ID:	78(LTS:226290, LTM:78)	Max. Points	1,00
Syllabus reference:	81.1.12.1 Ice and other contaminants	Points	1,00

23

Which of these statements about critical mach.No. is correct?

- ☐ As speed increases above critical mach No., parasite drag decreases rapidly.
- ☐ Flight at any speed above critical mach no. causes severe vibration of the aeroplane.
- ☒ Shock waves cannot occur at speeds below critical mach no.
- ☐ critical mach. No. is always greater than 1.

ID:	36(LTS:226248, LTM:36)	Max. Points	1,00
Syllabus reference:	81.2.2 Shock waves	Points	1,00

24

Which of these statements about an oblique shock wave are correct or incorrect?

I. The static temperature behind an oblique shock wave is lower than in front of it.
II. The static pressure behind an oblique shock wave is higher than in front of it.

- [x] I is incorrect, II is correct.
- [ ] I is incorrect, II is incorrect.
- [ ] I is correct, II is correct.
- [ ] I is correct, II is incorrect.

ID:	59(LTS:226271, LTM:59)	Max. Points	1,00
Syllabus reference:	81.2.2.2 Oblique shock waves	Points	1,00

25

During which type of stall does the angle of attack have the smallest value?

- ☐ Accelerated stall.
- ☐ Deep stall.
- ☐ Low speed stall.
- ☒ Shock stall.

ID:	18(LTS:226230, LTM:18)	Max. Points	1,00
Syllabus reference:	81.2.3.2 Effect on lift	Points	1,00

26

What is the effect of exceeding critical mach. No. on the stick force stability of an aeroplane with swept-back wings without any form of stability augmentation?

☒ **An increase, due to shock wave formation in the wing root area.**

☐ **No effect, becausecritical mach.No. is not relevant when considering stick force stability.**

☐ **A decrease, due to loss of lift in the wing root area.**

☐ **No effect, because stick force stability is independent of Mach number.**

ID:	42(LTS:226253, LTM:42)	Max. Points	1,00
Syllabus reference:	81.2.3.4 Effect on pitching moment	Points	0,00

27

What is the significance of the maximum allowed cruising altitude, based on the 1.3 g margin?

- ☐ **exceeding a load factor of 1.3 will cause permanent deformation of this aeroplane.**
- ☒ **a manoeuvre with a load factor of 1.3 will cause buffet onset.**
- ☐ **a manoeuvre with a load factor of 1.3 will cause a Mach number at which accelerated low speed stall occurs.**
- ☐ **a manoeuvre with a load factor of 1.3 will cause Mcrit to be exceeded.**

ID:	45(LTS:226256, LTM:45)	Max. Points	1,00
Syllabus reference:	81.2.4 Buffet onset	Points	1,00

28

Which of the following statements about static longitudinal stability is correct?

I. A requirement for positive static longitudinal stability of an aeroplane is, that the neutral point is behind the centre of gravity.
II. A wing with positive camber provides a positive contribution to static longitudinal stability, when the centre of gravity of the aeroplane is in front of the aerodynamic centre of the wing.

- [x] I is correct, II is correct.
- [ ] I is correct, II is incorrect.
- [ ] I is incorrect, II is incorrect.
- [ ] I is incorrect, II is correct.

ID: 87(LTS:226299, LTM:87)
Syllabus reference: 81.4.3.3 Neutral point
Max. Points 1,00
Points 1,00

29

Which of the following provides a positive contribution to static directional stability?

- ☐ A forward swept wing.
- ☐ A low wing as compared with a high wing.
- ☒ A dorsal fin.
- ☐ Moving the centre of gravity aft.

ID:	15(LTS:226227, LTM:15)	Max. Points	1,00
Syllabus reference:	81.4.4.4 Factors affecting static directional stability	Points	1,00

30

Sweepback of a wing positively influences:

1. static longitudinal stability.
2. static lateral stability.
3. dynamic longitudinal stability.

The combination that regroups all of the correct statements is:

- ☐ 3.
- ☐ 1.
- ☐ 2.
- ☒ 1, 2, 3.

ID: 5(LTS:226217, LTM:5)
Syllabus reference: 81.4.5.5 Factors affecting static lateral stability

Max. Points 1,00
Points 0,00

31

Which statement is about CG limits is correct?

- ☐ If the aft CG limit is correctly chosen, the forward CG limit is automatically determined as well.
- ☐ The forward CG limit is determined by stability considerations only.
- ☒ The forward CG limit is mainly determined by the amount of pitch control available from the elevator.
- ☐ The aft CG limit is determined by the maximum elevator deflection available.

ID:	21(LTS:226233, LTM:21)	Max. Points	1,00
Syllabus reference:	81.5.2.4 Location of centre of gravity	Points	1,00

32

The most important factor determining the required position of the Trimmable Horizontal Stabiliser (THS) for take off is the:

- [x] **position of the aeroplane's centre of gravity.**
- [ ] **centre of gravity position of the fuel.**
- [ ] **stall speed.**
- [ ] **total mass of the aeroplane.**

ID:	77(LTS:226289, LTM:77)	Max. Points	1,00
Syllabus reference:	81.5.8.3 Stabiliser trim	Points	1,00

33

Which of these statements about flutter are correct or incorrect?

I. If flutter occurs, IAS should be reduced.
II. Resistance to flutter increases with increasing wing stiffness.

- [x] **I is correct, II is correct.**
- [ ] **I is incorrect, II is incorrect.**
- [ ] **I is incorrect, II is correct.**
- [ ] **I is correct, II is incorrect.**

ID:	6(LTS:226218, LTM:6)	Max. Points	1,00
Syllabus reference:	81.6.1.1 Flutter	Points	1,00

34

The angle of attack of a fixed pitch propeller blade increases when:

- ☐ **velocity and RPM increase.**
- ☐ **velocity and RPM decrease.**
- ☒ **RPM increases and forward velocity decreases.**
- ☐ **forward velocity increases and RPM decreases.**

ID:	50(LTS:226262, LTM:50)	Max. Points	1,00
Syllabus reference:	81.7.1.3 Fixed and variable pitch/constant speed	Points	1,00

35

Which statement is correct when comparing a fixed pitch propeller with a constant speed propeller?

I. A constant speed propeller reduces fuel consumption over a range of cruise speeds.
II. A coarse fixed pitch propeller is more efficient during take-off.

- ☐ I is incorrect, II is incorrect.
- ☐ I is correct, II is correct.
- ☐ I is incorrect, II is correct.
- ☒ I is correct, II is incorrect

ID:	73(LTS:226285, LTM:73)	Max. Points	1,00
Syllabus reference:	81.7.1.4 Propeller efficiency versus speed	Points	1,00

36

Which statement is correct regarding the gyroscopic effect of a clockwise rotating propeller on a single engine aeroplane?

I. Pitch up produces left yaw.
II. Right yaw produces pitch up.

- [x] **I is correct, II is incorrect.**
- [ ] **I is correct, II is correct.**
- [ ] **I is incorrect, II is correct.**
- [ ] **I is incorrect, II is incorrect.**

ID:	8(LTS:226220, LTM:8)	Max. Points	1,00
Syllabus reference:	81.7.4.2 Gyroscopic precession	Points	0,00

37

**An aeroplane climbs to cruising level with a constant pitch attitude and maximum climb thrust, (assume no supercharger).
How do the following variables change during the climb? (Gamma = flight path angle)**

- [ ] **Gamma decreases, angle of attack increases, IAS remains constant.**
- [ ] **Gamma remains constant, angle of attack remains constant, IAS decreases.**
- [ ] **Gamma decreases, angle of attack remains constant, IAS decreases.**
- [x] **Gamma decreases, angle of attack increases, IAS decreases.**

ID:	24(LTS:226235, LTM:24)	Max. Points	1,00
Syllabus reference:	81.8.1.2 Straight steady climb	Points	1,00

38

What factors determine the distance travelled over ground of an aeroplane in a glide from a given altitude?

- [ ] The wind and CLMAX.
- [ ] The wind and the aeroplane's mass.
- [x] The wind and the lift/drag ratio.
- [ ] The wind and weight together with power loading, which is the ratio of power output to the weight.

ID:	31(LTS:226243, LTM:31)	Max. Points	1,00
Syllabus reference:	81.8.1.4 Straight steady glide	Points	1,00

39

An aeroplane performs a steady horizontal, co-ordinated turn with 45 degrees of bank at 230 kt TAS. The same aeroplane with the same bank angle and speed, but at a higher mass:

- ☐ will turn with a larger turn radius.
- ☐ will have a higher rate of turn.
- ☐ will turn with a smaller turn radius.
- ☒ will turn with the same radius, but might stall.

ID:	68(LTS:226280, LTM:68)	Max. Points	1,00
Syllabus reference:	81.8.1.5 Steady coordinated turn	Points	1,00

40

(Please view annex 081-0007 issue date October 2005)
The point in the diagram giving the lowest speed in unaccelerated flight is:

See Attachments:
1. 081-0007.gif Annex (Attachment No.1)

- [x] **point 4.**
- [ ] **point 2.**
- [ ] **point 3.**
- [ ] **point 1.**

ID: 92(LTS:226304, LTM:92)
Syllabus reference: 81.8.3 Particular points on the polar curve

Max. Points 1,00
Points 1,00

Attachment No. 1

Questions 40

Annex

081-0007.gif